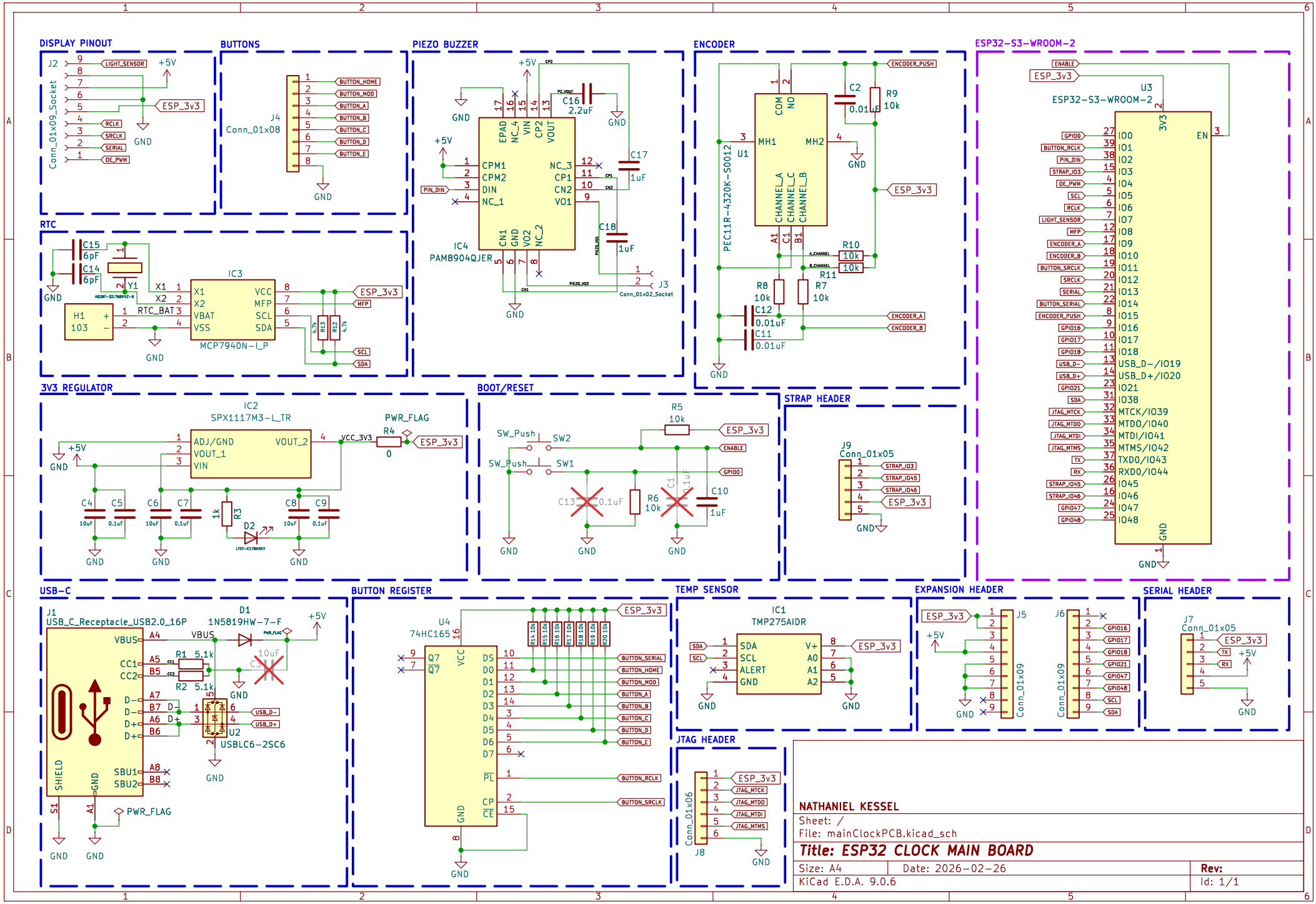
DISPLAY PINOUT
J2
Conn_01x09_Socket
LIGHT_SENSOR
RCLK
SRCLK
SERIAL
OE_PWM
+5V
ESP_3v3
GND
BUTTONS
J4
Conn_01x08
BUTTON_HOME
BUTTON_MOD
BUTTON_A
BUTTON_B
BUTTON_C
BUTTON_D
BUTTON_E
PIEZO BUZZER
IC4
PAM8904QJER
C16
2.2uF
C17
1uF
C18
1uF
J3
Conn_01x02_Socket
PIN_DIN
ENCODER
U1
PEC11R-4320K-S0012
C2
0.01uF
R9
10k
R10
10k
R11
10k
R8
10k
R7
10k
C12
0.01uF
C11
0.01uF
ENCODER_PUSH
ENCODER_A
ENCODER_B
ESP32-S3-WROOM-2
U3
ESP32-S3-WROOM-2
ENABLE
RTC
IC3
MCP7940N-I_P
C15
6pF
C14
6pF
Y1
H1
103
RTC_BAT
MFP
SCL
SDA
3V3 REGULATOR
IC2
SPX1117M3-L_TR
PWR_FLAG
R4
0
VCC_3V3
C4
10uF
C5
0.1uF
C6
10uF
C7
0.1uF
R3
1k
D2
C8
10uF
C9
0.1uF
BOOT/RESET
SW_Push
SW2
SW1
R5
10k
R6
10k
C13
0.1uF
C1
0.1uF
C10
1uF
STRAP HEADER
J9
Conn_01x05
STRAP_IO3
STRAP_IO45
STRAP_IO46
USB-C
J1
USB_C_Receptacle_USB2.0_16P
D1
1N5819HW-7-F
R1
5.1k
R2
5.1k
C3
10uF
U2
USBLC6-2SC6
USB_D-
USB_D+
SHIELD
S1
A1
PWR_FLAG
BUTTON REGISTER
U4
74HC165
BUTTON_SERIAL
BUTTON_HOME
BUTTON_MOD
BUTTON_A
BUTTON_B
BUTTON_C
BUTTON_D
BUTTON_E
BUTTON_RCLK
BUTTON_SRCLK
TEMP SENSOR
IC1
TMP275AIDR
SDA
SCL
ALERT
EXPANSION HEADER
J5
J6
Conn_01x09
SERIAL HEADER
J7
Conn_01x05
TX
RX
JTAG HEADER
J8
Conn_01x06
JTAG_MTCK
JTAG_MTDO
JTAG_MTDI
JTAG_MTMS
NATHANIEL KESSEL
Sheet: /
File: mainClockPCB.kicad_sch
Title: ESP32 CLOCK MAIN BOARD
Size: A4
Date: 2026-02-26
Rev:
KiCad E.D.A. 9.0.6
Id: 1/1